

# FML Assignment 4

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## R Markdown

### Loading necessary libraries

```
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
library(e1071)  
library(tinytex)  
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      intersect, setdiff, setequal, union
```

### Loading the dataset

```
Heart.disease <- read.csv("C:\\Users\\nisha\\Downloads\\Heart_disease.csv")
```

### Inspecting the first few rows of the dataset to understand its structure

```
head(Heart.disease)
```

```
##   Age Sex chest_pain_type Blood_Pressure Cholestrol Fasting_Blood_Sugar
## 1  63  1              0             145         233              1
## 2  37  1              1             130         250              0
## 3  41  0              1             130         204              0
## 4  56  1              1             120         236              0
## 5  57  0              0             120         354              0
## 6  57  1              0             140         192              0
##   Rest_ECG MAX_HeartRate Exercise
## 1         0          150         0
## 2         1          187         0
## 3         0          172         0
## 4         1          178         0
## 5         1          163         1
## 6         1          148         0
```

## Interpretatoin

The above code shows the first six records of data, giving a general overview of the type of data and parameters such as age, gender, chest pain type, blood pressure, cholesterol.

## Creating the “Target” dummy variable based on MAX\_HeartRate

```
Heart.disease <- Heart.disease %>%
  mutate(Target = factor(ifelse(MAX_HeartRate > 170, "Yes", "No")))
```

## Interpretation

The following code adds a new category variable to the Heart\_disease collection named Target. It labels ‘Yes’ if MAX\_HeartRate reaches 170, and ‘No’ otherwise, suggesting a high risk of developing heart disease.

## Creating the ‘BP\_New’ dummy variable

```
Heart.disease <- Heart.disease %>%
  mutate(BP_New = factor(ifelse(Blood_Pressure > 120, "Yes", "No")))
```

## Interpretation

This code generates a new categorical variable known as BP\_New in the data-set of . It labels “Yes” if Blood\_Pressure>120 and “No” otherwise meaning high blood pressure is present.

## Checking the levels of the Target variable

```
print(levels(Heart.disease$Target))
```

```
## [1] "No" "Yes"
```

## Baseline prediction: Most frequent class in the dataset

```
Target_table <- table(Heart.disease$Target)
print(Target_table)
```

```
##
## No Yes
## 245  58
```

## Interpretation

This creates a Target\_table that contains the amount of times any of the levels in the Target variable are repeated. This will show us the number of individuals who are having a higher risk of heart disease of “Yes” compared to those labeled as “No”.

## Prediction should be the majority class

```
baseline_prediction <- names(Target_table)[which.max(Target_table)]
print(paste("Baseline prediction for heart disease:", baseline_prediction))
```

```
## [1] "Baseline prediction for heart disease: No"
```

## Select the first 30 records

```
Heart.disease30 <- Heart.disease[1:30, c("Target", "BP_New", "chest_pain_type")]
```

## Interpretation

This generates a new dataframe called Heart\_disease30 and extracts the first 30 data points from the dataset. It specifically selects three columns: ‘Target’, ‘BP\_New’, and ‘chest\_pain\_type’. This subset will incorporate further analyses of these variables, such as determining the likelihood of an element in relation to the illness prognosis of the human heart.

## Creating Object 1 and Object 2, along with printing Object 1 and 2

```
Object1 <- ftable(Heart.disease30)
print(Object1)
```

```
##           chest_pain_type 0 1
## Target BP_New
## No      No              2 2
##         Yes             7 8
## Yes     No              0 3
##         Yes             3 5
```

### Interpretation

The Object1 is a frequency table for the first 30 records of Heart\_disease presenting the count of the variability of “Target,” “BP\_New,” and “chest\_pain\_type.” This table shows an association between presence of heart disease and the predictors.

```
Object2 <- ftable(Heart.disease30[, -1])
print(Object2)
```

```
##           chest_pain_type 0 1
## BP_New
## No              2 5
## Yes            10 13
```

### Interpretation

This generates a frequency table for the basic Object 2 that only includes the first 30 records of the dataset, including the variables “BP\_New” and “chest\_pain\_type”, but completely ignores the “Target” variable. This table depicts the raw frequencies of various versions of these two predictors, which is valuable for understanding them and their potential link.

## Calculate Probability Outcomes for the four combinations

```
P1 = Object1[3, 1] / Object2[1, 1]
P2 = Object1[4, 1] / Object2[2, 1]
P3 = Object1[3, 2] / Object2[1, 2]
P4 = Object1[4, 2] / Object2[2, 2]
```

## Interpretation

This code computes four probabilities depending on the frequency of combinations in the dataset:

P1: Probability of heart disease (“Yes”), blood pressure (“No”), and chest pain type 0. P2: Probability of no heart disease (“No”) when blood pressure is “Yes” and chest discomfort type is 0. P3: Probability of not having heart disease with blood pressure “No” and chest discomfort type 1. P4: Heart disease risk with blood pressure “Yes” and chest discomfort type 1. # Display calculated probabilities for the four combinations

## Probability calculations for classification

```
Probability_Target <- rep(0, 30)
for (i in 1:30) {
  if (Heart.disease30$BP_New[i] == "Yes") {
    if (Heart.disease30$chest_pain_type[i] == 0) {
      Probability_Target[i] = P1
    } else if (Heart.disease30$chest_pain_type[i] == 1) {
      Probability_Target[i] = P3
    }
  } else {
    if (Heart.disease30$chest_pain_type[i] == 0) {
      Probability_Target[i] = P2
    } else if (Heart.disease30$chest_pain_type[i] == 1) {
      Probability_Target[i] = P4
    }
  }
}

Heart.disease30$Probability_Target <- Probability_Target
Heart.disease30$Pred_Probability <- ifelse(Heart.disease30$Probability_Target > 0.5, "Yes", "No")
```

## Interpretation

In this code, a vector called Probability\_Target is defined to store a projected probability for each of the set's first 30 records. The likelihood of heart disease was calculated for each record using the values obtained from BP\_New and chest\_pain\_type, and the loop control variable was increased. Probability\_Target is assigned the relevant probabilities P1, P2, P3, and P4 based on these circumstances. Finally, a new column called Pred\_Probability is created, and entries are sorted as “Yes” if the likelihood is greater than 0.5, otherwise “No”.

## Print the predictions for the first 30 records

```
print(Heart.disease30[, c("Target", "Pred_Probability")])
```

```
##      Target Pred_Probability
## 1      No                No
## 2     Yes                Yes
```

|       |     |     |
|-------|-----|-----|
| ## 3  | Yes | Yes |
| ## 4  | Yes | No  |
| ## 5  | No  | No  |
| ## 6  | No  | No  |
| ## 7  | No  | Yes |
| ## 8  | Yes | No  |
| ## 9  | No  | Yes |
| ## 10 | Yes | Yes |
| ## 11 | No  | No  |
| ## 12 | No  | Yes |
| ## 13 | Yes | Yes |
| ## 14 | No  | No  |
| ## 15 | No  | No  |
| ## 16 | No  | No  |
| ## 17 | Yes | No  |
| ## 18 | No  | No  |
| ## 19 | Yes | No  |
| ## 20 | No  | No  |
| ## 21 | No  | No  |
| ## 22 | Yes | Yes |
| ## 23 | Yes | No  |
| ## 24 | No  | Yes |
| ## 25 | Yes | No  |
| ## 26 | No  | Yes |
| ## 27 | No  | Yes |
| ## 28 | No  | No  |
| ## 29 | No  | Yes |
| ## 30 | No  | Yes |

## Interpretation

This outputs the number of real targets intended at heart disease rates as well as the anticipated likelihood for the first 30 records in the data set. This comparison is important to examine how well the model is doing to forecast a particular event when compared to the actual outcomes

## — Q2c: Manual Calculation of Naive Bayes Probability —

Manual Naive Bayes calculation for BP\_New = Yes and chest\_pain\_type = 1

```
count_bp_yes_cp_1 <- nrow(Heart.disease30 %>% filter(BP_New == "Yes", chest_pain_type == 1))
count_bp_yes <- nrow(Heart.disease30 %>% filter(BP_New == "Yes"))
count_cp_1 <- nrow(Heart.disease30 %>% filter(chest_pain_type == 1))
count_total <- nrow(Heart.disease30)
```

## Interpretation

This will count the following among the dataset's first 30 rows.

count\_bp\_yes\_cp\_1: The number of cases in which blood pressure is “Yes” and the kind of chest pain is 1. count\_bp\_yes: The total number of occasions where blood pressure is “Yes”. count\_cp\_1: The total number of occasions when the chest pain type is one. count\_total: The total number of records (30) being examined.

## Prior probabilities

```
prior_target_yes <- sum(Heart.disease30$Target == "Yes") / count_total
prior_bp_yes <- count_bp_yes / count_total
prior_cp_1 <- count_cp_1 / count_total
```

## Likelihood (P(chest\_pain\_type=1 | BP\_New=Yes))

```
likelihood_cp_1_given_bp_yes <- count_bp_yes_cp_1 / count_bp_yes
```

## Posterior (Bayes rule)

```
posterior_prob <- (likelihood_cp_1_given_bp_yes * prior_target_yes) / prior_bp_yes
print(paste("Manual Naive Bayes probability for BP_New = Yes and chest_pain_type = 1:", posterior_prob))
```

```
## [1] "Manual Naive Bayes probability for BP_New = Yes and chest_pain_type = 1: 0.270321361058601"
```

## Interpretation

This code assesses the risk of having heart disease if one's blood pressure is high (“Yes”) and the chest pain kind is 1. It combines the likelihood of experiencing chest discomfort type 1 with high blood pressure and the overall likelihood of having a cardiac condition. Finally, the probability indicates the chance of having heart disease under these settings.

## — Q3: Full Dataset Analysis —

## Splitting the data into 60% training and 40% validation

```
set.seed(123) # for reproducibility
train.index <- sample(row.names(Heart.disease), 0.6 * dim(Heart.disease)[1])
valid.index <- setdiff(row.names(Heart.disease), train.index)
train.df <- Heart.disease[train.index, ]
valid.df <- Heart.disease[valid.index, ]
```

## Interpretation

This code divides the dataset into two parts, 60% for training and 40% for validation. Creates a training set using the sample function and the remaining indices for the model's validation set.

## Train Naive Bayes Model

```
nb_model <- naiveBayes(Target ~ chest_pain_type + BP_New, data = train.df)
```

## Interpretation

This line of code builds a Naive Bayes classifier (nb\_model) based on the training data included in the variable train.df with Target variable which will mean heart disease pointing at chest\_pain\_type and BP\_New. They make a clear connection of these predictors to the target outcome in order to classify them.

## Predictions for training set

```
train_pred <- predict(nb_model, train.df)
```

## Convert train\_pred to factor with levels

```
train_pred <- factor(train_pred, levels = levels(Heart.disease$Target))
```

## Confusion Matrix for training set

```
conf_matrix1 <- confusionMatrix(train_pred, train.df$Target, positive = 'Yes')
print(conf_matrix1)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##           No 149 32
##           Yes  0  0
##
##           Accuracy : 0.8232
##           95% CI : (0.7597, 0.8758)
##           No Information Rate : 0.8232
##           P-Value [Acc > NIR] : 0.5471
##
```

```
##                Kappa : 0
##
## Mcnemar's Test P-Value : 4.251e-08
##
##          Sensitivity : 0.0000
##          Specificity : 1.0000
##          Pos Pred Value :    NaN
##          Neg Pred Value : 0.8232
##          Prevalence : 0.1768
##          Detection Rate : 0.0000
##          Detection Prevalence : 0.0000
##          Balanced Accuracy : 0.5000
##
##          'Positive' Class : Yes
##
```

## Interpretation

This computes the confusion matrix, `conf_matrix1`, of the train predictions, `train_pred`, based on the site labels in `train.df`. This matrix outlines outcomes of positives that are true and false positives, and negatives that are true and false negatives, by assessing accuracy and other parameters, mainly the positive category or 'Yes.'

## Validate the model on the validation set

```
valid_pred <- predict(nb_model, valid.df)
valid_pred <- factor(valid_pred, levels = levels(Heart.disease$Target))
```

## Confusion Matrix for validation set

```
conf_matrix_valid <- confusionMatrix(valid_pred, valid.df$Target, positive = 'Yes')
print(conf_matrix_valid)
```

```
## Confusion Matrix and Statistics
##
##          Reference
## Prediction No Yes
##          No  96  26
##          Yes   0   0
##
##          Accuracy : 0.7869
##          95% CI : (0.7035, 0.8558)
##          No Information Rate : 0.7869
##          P-Value [Acc > NIR] : 0.5523
##
##          Kappa : 0
```

```
##
## McNemar's Test P-Value : 9.443e-07
##
##      Sensitivity : 0.0000
##      Specificity : 1.0000
##      Pos Pred Value :    NaN
##      Neg Pred Value : 0.7869
##      Prevalence : 0.2131
##      Detection Rate : 0.0000
##      Detection Prevalence : 0.0000
##      Balanced Accuracy : 0.5000
##
##      'Positive' Class : Yes
##
```

## Interpretation

The validation predictions' confusion matrix on variable `valid_pred` compared to the real target variable in `valid.df`. This matrix displays the numerical outcomes of the model applied to the unseen data, assessing the proportion of very accurate positive, negative, and wrong predictions. It enables assessing the model's effectiveness in terms of ranking complications, accuracy, precision, memory, etc., which can provide insight into the viability and quality of diagnosing cardiac sickness.